

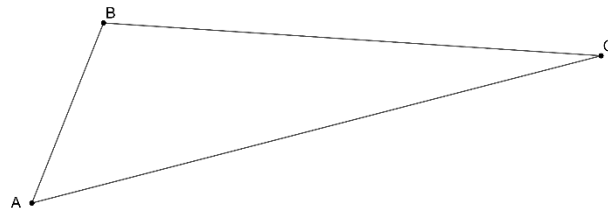
Geometry
Unit 11
Lesson 6 Homework

Name **Answer Key**
Date _____
Period _____

1. $\triangle ABC$ is shown, where $AB = 4$, $BC = 8$, and $AC = 10$.

List the angles in order from least to greatest.

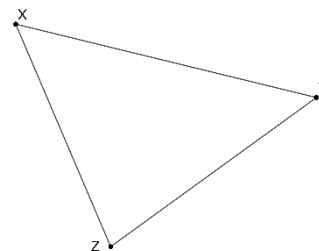
$\angle C, \angle A, \angle B$



$\triangle XYZ$ is shown. Use the triangle to complete questions 2 through 4.

2. Use a protractor to measure each of the angles.

$m\angle Z = 76^\circ$ $m\angle X = 53^\circ$ $m\angle Y = 51^\circ$



3. List the angles in order from smallest to largest.

$\angle Y, \angle X, \angle Z$

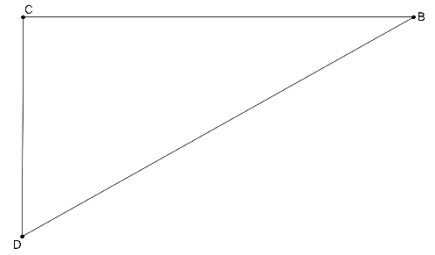
4. List the side lengths in order from smallest to largest.

$\overline{XZ}, \overline{YZ}, \overline{XY}$

5. $\triangle BCD$ is shown, where $m\angle B = 30^\circ$, $m\angle D = 60^\circ$, and $m\angle C = 90^\circ$.

List the side lengths in order from largest to smallest.

$\overline{BD}, \overline{BC}, \overline{CD}$



For questions 6 through 8, determine whether the given side lengths can form a triangle.

6. 12 cm, 14 cm, and 20 cm

$12 + 14 = 26; 26 > 20$ Yes it is a triangle.

7. 2 in, 3 in, and 5 in

$2 + 3 = 5; 5 = 5$ No, cannot be a triangle.

8. 3 ft, 8 ft, and 12 ft

$3 + 8 = 11; 11 < 12$ No, cannot be a triangle.

For questions 9 and 10, two sides of a triangle are given. Determine a possible side length for the third side.

9. 7 in and 5 in

$7 + 5 = 12, 7 - 5 = 2$ Any value greater than 2 and less than 12.

10. 10 m and 2 m

$10 + 2 = 12, 10 - 2 = 8$ Any value greater than 8 and less than 12.